Multi-Step Linear Equations with Rational Coefficients

Clearing fractions with the LCD, equations with the variable on both sides, and inequalities with rational coefficients

Directions.

- Show every step. A step is legal only if you do the same thing to *both* sides of the equation.
- When fractions appear, multiplying both sides by the least common denominator (LCD) clears them all at once — multiply *every* term, including the terms with no fraction.
- Give exact answers (a fraction, not a rounded decimal), and check your solution in the original equation.
- In an inequality, multiplying or dividing both sides by a **negative** number reverses the inequality symbol.

1. Solve for x. $\dfrac{x}{3} + 5 = 9$

Answer: ________________

2. Solve for x. $\dfrac{2}{5}x - 3 = 7$

Answer: ________________

3. Solve for x. Multiply both sides by the LCD first. $\frac{x}{4} + \frac{x}{6} = 10$

Answer: ____________________

4. Solve for x. $\frac{3}{4}(x - 8) = 6$

Answer: ____________________

5. Solve for x. The variable appears on both sides. $\frac{2}{3}x + 1 = \frac{x}{6} + 4$

Answer: ____________________

6. Solve the inequality and write the solution in symbols. $\dfrac{x}{2} - 3 > 4$

Answer: ________________

7. Solve for x. Watch the binomial numerators when you multiply by the LCD.

$$\frac{x+2}{3} + \frac{x-1}{2} = 6$$

Answer: ________________

8. Solve for x.

$$\frac{3}{4}x - \frac{2}{3} = \frac{x}{6} + \frac{5}{3}$$

Answer: ____________________

9. Solve the inequality and write the solution in symbols. Decide carefully whether the symbol turns around. $-\frac{2}{3}x + 1 \geq 5$

Answer: ____________________

10. Clear the fractions from the equation below by multiplying *both sides* by the LCD, then write the equivalent equation that has only integer coefficients. (Do not solve it.)

$$\frac{5}{6}x - \frac{1}{2} = \frac{1}{3}x + 2$$

Answer: __________________

11. Solve for x. Distribute the outside factor before you collect like terms.

$$\frac{1}{2}\left(\frac{x}{3} - 4\right) + \frac{5}{6}x = 3$$

Answer: __________________

12. Solve for x. Multiply by the LCD of all three fractions at once, then finish.

$$\frac{3(2x-1)}{4} - \frac{2(x+1)}{3} = \frac{5}{12}x + 4$$

Answer: __________________